# Equivalent Fractions and Mixed Numbers

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**KEY WORDS** numerator *(taurunga)* · denominator *(tauraro)* · equivalent *(ōrite)* · simplify *(whakamāmā)*

## A Equivalent fractions ●○○

**Remember:** Multiply or divide the top and the bottom by the same number. The value of the fraction does not change.

**1.** Fill in the box to make the fractions equal.

**a)** $\frac{2}{3} = \frac{\square}{12}$ **b)** $\frac{3}{5} = \frac{\square}{15}$ **c)** $\frac{1}{4} = \frac{5}{\square}$ **d)** $\frac{5}{6} = \frac{\square}{12}$

**e)** $\frac{3}{8} = \frac{9}{\square}$ **f)** $\frac{4}{7} = \frac{\square}{35}$ **g)** $\frac{7}{10} = \frac{\square}{100}$ **h)** $\frac{5}{9} = \frac{20}{\square}$

**i)** $\frac{1}{2} = \frac{\square}{18}$ **j)** $\frac{2}{9} = \frac{12}{\square}$ **k)** $\frac{11}{12} = \frac{\square}{36}$ **l)** $\frac{3}{4} = \frac{\square}{100}$

## B Simplifying fractions ●○○

**Remember:** Divide the top and the bottom by their highest common factor.

**2.** Write each fraction in its simplest form.

**a)** $\frac{6}{8}$ = _____ **b)** $\frac{10}{15}$ = _____ **c)** $\frac{12}{18}$ = _____ **d)** $\frac{9}{21}$ = _____

**e)** $\frac{20}{28}$ = _____ **f)** $\frac{16}{40}$ = _____ **g)** $\frac{27}{45}$ = _____ **h)** $\frac{36}{60}$ = _____

**i)** $\frac{35}{50}$ = _____ **j)** $\frac{22}{55}$ = _____ **k)** $\frac{42}{56}$ = _____ **l)** $\frac{75}{100}$ = _____

## C Improper fractions to mixed numbers ●●○

**Remember:** Divide the top by the bottom. The remainder becomes the top of the fraction part.

**3.** Write each improper fraction as a mixed number.

**a)** $\frac{7}{2}$ = _____ **b)** $\frac{11}{3}$ = _____ **c)** $\frac{17}{5}$ = _____ **d)** $\frac{9}{4}$ = _____

**e)** $\frac{23}{6}$ = _____ **f)** $\frac{29}{8}$ = _____ **g)** $\frac{31}{10}$ = _____ **h)** $\frac{40}{7}$ = _____

**i)** $\frac{19}{3}$ = _____ **j)** $\frac{50}{9}$ = _____ **k)** $\frac{47}{12}$ = _____ **l)** $\frac{38}{5}$ = _____



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## D Mixed numbers to improper fractions ●●○

**Remember:** Multiply the whole number by the bottom, then add the top.

**4.** Write each mixed number as an improper fraction.

**a)** $2\frac{1}{3}$ = _____ **b)** $1\frac{3}{4}$ = _____ **c)** $3\frac{2}{5}$ = _____ **d)** $4\frac{1}{2}$ = _____

**e)** $2\frac{5}{6}$ = _____ **f)** $5\frac{3}{8}$ = _____ **g)** $6\frac{2}{3}$ = _____ **h)** $3\frac{7}{10}$ = _____

**i)** $7\frac{1}{4}$ = _____ **j)** $4\frac{5}{9}$ = _____ **k)** $10\frac{1}{2}$ = _____ **l)** $2\frac{7}{12}$ = _____

## E Comparing fractions ●●●

**Remember:** Write both fractions with the same denominator, then compare the tops.

**5.** Which fraction is larger?

**a)** $\frac{2}{3}$ or $\frac{3}{5}$ **b)** $\frac{3}{4}$ or $\frac{5}{8}$ **c)** $\frac{1}{2}$ or $\frac{4}{9}$

**d)** $\frac{5}{6}$ or $\frac{7}{9}$ **e)** $\frac{3}{10}$ or $\frac{1}{4}$ **f)** $\frac{7}{12}$ or $\frac{5}{8}$

**g)** $\frac{4}{5}$ or $\frac{7}{9}$ **h)** $\frac{2}{7}$ or $\frac{1}{3}$ **i)** $\frac{11}{15}$ or $\frac{7}{10}$

**6.** Write each set of fractions in order, smallest first.

**a)** $\frac{1}{2}, \frac{3}{8}, \frac{3}{4}$ **b)** $\frac{2}{3}, \frac{5}{6}, \frac{7}{12}$ **c)** $\frac{3}{5}, \frac{1}{2}, \frac{7}{10}$

**d)** $\frac{2}{3}, \frac{3}{4}, \frac{5}{8}$ **e)** $\frac{1}{3}, \frac{2}{9}, \frac{5}{18}$ **f)** $\frac{3}{4}, \frac{4}{5}, \frac{7}{10}$

## ★ Extension

**7.** Write $\frac{84}{126}$ in its simplest form.

**8.** Write a fraction that is greater than $\frac{1}{3}$ and less than $\frac{1}{2}$.



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# Equivalent Fractions and Mixed Numbers

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**KEY WORDS** numerator *(taurunga)* · denominator *(tauraro)* · equivalent *(ōrite)* · simplify *(whakamāmā)*

## A Equivalent fractions ●○○

**Remember:** Multiply or divide the top and the bottom by the same number. The value of the fraction does not change.

**1.** Fill in the box to make the fractions equal.

**a)** $\frac{3}{4} = \frac{\square}{12}$ **b)** $\frac{2}{5} = \frac{\square}{20}$ **c)** $\frac{1}{6} = \frac{5}{\square}$ **d)** $\frac{5}{8} = \frac{\square}{16}$

**e)** $\frac{2}{7} = \frac{6}{\square}$ **f)** $\frac{4}{9} = \frac{\square}{45}$ **g)** $\frac{3}{10} = \frac{\square}{100}$ **h)** $\frac{5}{6} = \frac{20}{\square}$

**i)** $\frac{1}{2} = \frac{\square}{14}$ **j)** $\frac{4}{9} = \frac{24}{\square}$ **k)** $\frac{7}{12} = \frac{\square}{36}$ **l)** $\frac{1}{4} = \frac{\square}{100}$

## B Simplifying fractions ●○○

**Remember:** Divide the top and the bottom by their highest common factor.

**2.** Write each fraction in its simplest form.

**a)** $\frac{4}{6}$ = _____ **b)** $\frac{12}{15}$ = _____ **c)** $\frac{14}{21}$ = _____ **d)** $\frac{8}{20}$ = _____

**e)** $\frac{18}{30}$ = _____ **f)** $\frac{15}{40}$ = _____ **g)** $\frac{28}{49}$ = _____ **h)** $\frac{48}{72}$ = _____

**i)** $\frac{45}{60}$ = _____ **j)** $\frac{26}{65}$ = _____ **k)** $\frac{36}{54}$ = _____ **l)** $\frac{60}{100}$ = _____

## C Improper fractions to mixed numbers ●●○

**Remember:** Divide the top by the bottom. The remainder becomes the top of the fraction part.

**3.** Write each improper fraction as a mixed number.

**a)** $\frac{9}{2}$ = _____ **b)** $\frac{10}{3}$ = _____ **c)** $\frac{13}{5}$ = _____ **d)** $\frac{15}{4}$ = _____

**e)** $\frac{25}{6}$ = _____ **f)** $\frac{27}{8}$ = _____ **g)** $\frac{43}{10}$ = _____ **h)** $\frac{37}{7}$ = _____

**i)** $\frac{22}{3}$ = _____ **j)** $\frac{52}{9}$ = _____ **k)** $\frac{41}{12}$ = _____ **l)** $\frac{33}{5}$ = _____



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## D Mixed numbers to improper fractions ●●○

**Remember:** Multiply the whole number by the bottom, then add the top.

**4.** Write each mixed number as an improper fraction.

**a)** $3\frac{1}{2}$ = _____ **b)** $1\frac{2}{3}$ = _____ **c)** $2\frac{3}{5}$ = _____ **d)** $4\frac{3}{4}$ = _____

**e)** $3\frac{1}{6}$ = _____ **f)** $2\frac{5}{8}$ = _____ **g)** $5\frac{1}{3}$ = _____ **h)** $4\frac{9}{10}$ = _____

**i)** $6\frac{3}{4}$ = _____ **j)** $3\frac{4}{9}$ = _____ **k)** $9\frac{1}{2}$ = _____ **l)** $2\frac{5}{12}$ = _____

## E Comparing fractions ●●●

**Remember:** Write both fractions with the same denominator, then compare the tops.

**5.** Which fraction is larger?

**a)** $\frac{3}{4}$ or $\frac{2}{3}$ **b)** $\frac{2}{5}$ or $\frac{3}{10}$ **c)** $\frac{1}{2}$ or $\frac{5}{9}$

**d)** $\frac{5}{6}$ or $\frac{3}{4}$ **e)** $\frac{7}{10}$ or $\frac{3}{4}$ **f)** $\frac{5}{12}$ or $\frac{3}{8}$

**g)** $\frac{3}{5}$ or $\frac{5}{9}$ **h)** $\frac{3}{7}$ or $\frac{1}{2}$ **i)** $\frac{8}{15}$ or $\frac{3}{5}$

**6.** Write each set of fractions in order, smallest first.

**a)** $\frac{1}{2}$, $\frac{5}{8}$, $\frac{1}{4}$ **b)** $\frac{1}{3}$, $\frac{5}{6}$, $\frac{5}{12}$ **c)** $\frac{4}{5}$, $\frac{1}{2}$, $\frac{3}{10}$

**d)** $\frac{1}{3}$, $\frac{1}{4}$, $\frac{3}{8}$ **e)** $\frac{2}{3}$, $\frac{5}{9}$, $\frac{11}{18}$ **f)** $\frac{1}{4}$, $\frac{1}{5}$, $\frac{3}{10}$

## ★ Extension

**7.** Write $\frac{72}{120}$ in its simplest form.

**8.** Write a fraction that is greater than $\frac{1}{4}$ and less than $\frac{1}{3}$.


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